DEPARTMENT OF ELECTRICAL & COMPUTER ENGINEERING | FAMU-FSU College of Engineering

EEL 4911C - ECE Senior Design Project I

Team 309: RFID-Enabled Access Control Switch for Equipment Lockout

Document: Updated Design Concept and Detailed Design

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Introduction

Team 309 has iterated on several design concepts for an RFID-based access control system intended to regulate the use of lab equipment at FSU. This updated document outlines the current system architecture, and how the design aligns with relevant industry codes and standards. Ensuring safety and compliance is essential for both system integrity and user protection. The team's adherence to UL, NEC, FSU, and IEEE standards demonstrates a commitment to producing a secure and professional-grade solution.

Current Design

The design consists of two physical subsystems at each equipment station, supported by a centralized administrative interface. Each system will communicate with the main computer through the WIFI network available at FSU. The two objects will take the form of the image below. Exact specifications of the objects will not be discussed here, and for more information, refer to the Preliminary Detailed Design document. The following is a brief breakdown of each object.

Main unit housing – This unit contains the Arduino Uno WiFi Rev2, which was selected for its built-in wireless communication capabilities and compatibility with serial interfaces. This microcontroller is responsible for authenticating users by reading input from the PN532 RFID module, sending a request to the local SQL database via Wi-Fi, and controlling a solid-state relay (SSR) to enable or disable power to the connected equipment. The housing is a 3D-printed enclosure that is secured with a padlock for safety and tamper resistance. It is designed to allow maintenance access while preventing user interference. All equipment that will be protected can be integrated into this design without modifying it and can be removed from the Main unit housing at any time by way of a lock and key used to restrict maintenance access.

Auxiliary unit housing – This will be a separate unit from the Main unit housing and will hold an interface (LEDs) and an RFID reader. This will be connected to the Main unit housing and will be the piece of equipment that users interact with.

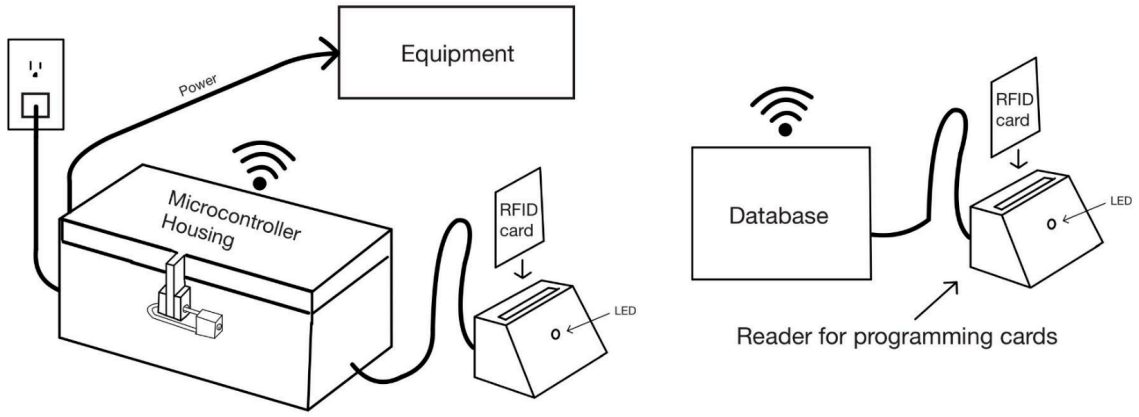
Main computer – The main logic controller in the design will be an Arduino Uno WIFI Rev2, which will be housed inside the main housing unit. This microcontroller will be connected to the LEDS, SSR, and RFID reader through its GPIO pins. The microcontroller has built-in wireless capabilities that will allow it to connect to the FSU wireless network and, by extension, the database unit.

Reader for programming cards – The PN532 RFID Reader will be a module connected to the Main computer that will be used to grant and revoke access to equipment by way of editing the database via the administrative unit. The administrative unit will update its ID and permissions when the RFID/NFC card is added and written to.

Administrative unit - Website that can be accessed by school administrators. Using an HTML/PHP-based website and a locally hosted database, this will serve as the master controller behind

the RFIDs. By using CRUD operations via the website, the administrator can control, add, or remove access to machines. This is hooked up to the reader, which will allow individual cards to be updated and edited.

This is an image of what the design will likely look like:



Codes and Standards

- UL certification

All electrical equipment that will be integrated into this design will have a UL certification. This certification comes from UL Solutions and shows that the equipment has quality assurance, meaning it has a high chance of working and not causing harm. This certification also means the equipment meets standards that specifically apply to that component.

- NEC

The National Electric Code governs all electrical work that is done in the USA. While this code intends to govern electrical installations, it has many standards that this design will adhere to. For example, 110.3B states that all equipment must be used in the manner it was designed for. This design will adhere to many such standards, only using equipment that has been tested for the applications that this design will encounter.

- FSU Codes

These codes have the most to do with the design, but at the same time have very little to do with it. This design will uphold all FSU codes and standards regarding IT data security standards (to the best of the ability of Team 309). This project is designed for use at FSU and thus must be appropriate for that application. It should be noted that the data stored on the microcontroller and the Main computer will be protected by location, either under a padlock or behind a door with restricted access. All data transmitted through a WIFI signal will be protected by FSU's network protocols.

- IEEE Codes

All team members of Team 309 will finally hold themselves and each other accountable to the IEEE Code of Ethics. This code is a vital part of the engineering community and sets guidelines that each individual must follow to ensure the success of all projects.

Public Safety and Other Factors

Public health, safety, and welfare are all major parts of this design, as the entire purpose of this project is to safeguard students from obtaining injuries on equipment they are untrained on. If this design further endangered students, it would be fundamentally flawed. To prevent such issues, Team 309 will adhere to all codes listed above. This will ensure the produced product is assembled safely and utilises tested parts. Further, all medium voltage power present in the project will be insulated except for where those wires terminate in the design. All wiring components will also be in an enclosed location, which will not be accessible to users. This project will also have a small global impact since all parts are man-made and need to be sourced from materials. This is not a direct concern for this design and will be addressed by bringing business to companies with reputable suppliers. While not the main focus of this project, Team 309 would like to have as little of an environmental impact as possible.

As for other aspects that were considered, the project is designed specifically for US-based academic institutions, aligning with standard Western interface norms and English-language support. Second, the economic impact of this project was considered throughout the design process. The design is built to be modular and be used across many systems at FSU. This means that each unit will be cost-efficient and scalable. Finally, and most importantly, usability was one of the main focuses of this project. The intent of the design is to be used on all lab equipment at FSU; thus, making it simple and easy to use was a critical part of this design. The team acted as experts on this topic since all members are students. As the representative end-users, we discussed the simplest and most optimal way to implement this system, and included it in the design.

Conclusion

Team 309 has developed a comprehensive, standards-compliant design for an RFID-enabled access control system intended to improve safety and accountability in student laboratories. The design separates system functionality into three key components: a locked main unit for control logic and power switching, an auxiliary user interface unit with RFID scanning and visual feedback, and a host computer running a local SQL database and administrative interface.

All components were selected based on functionality, ease of integration, cost-effectiveness, and compliance with safety and electrical standards such as UL and NEC guidelines. The team has also prioritized coding best practices, maintainability, and modularity to ensure the system can be expanded or replicated across additional lab stations in the future.

With design finalization complete, Team 309 is prepared to transition into the implementation and testing phase at the start of the Spring semester. The remaining work includes assembling hardware components, developing and verifying embedded software, integrating database communications, and performing system-level testing.